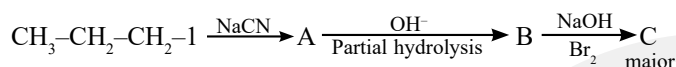


CHAPTER 10

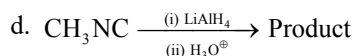
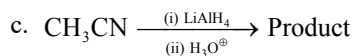
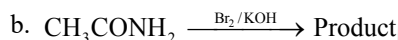
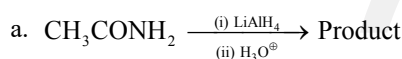
Amines

Preparation of Amines

1. Identify the major product C formed in the following reaction sequence: (2024)



- a. butanamide
b. α -bromobutanoic acid
c. propylamine
d. butylamine
2. Which of the following reactions will NOT give primary amine as the product? (2023)



3. Match List - I with List - II: (2022 Re)

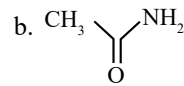
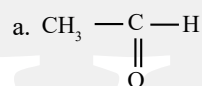
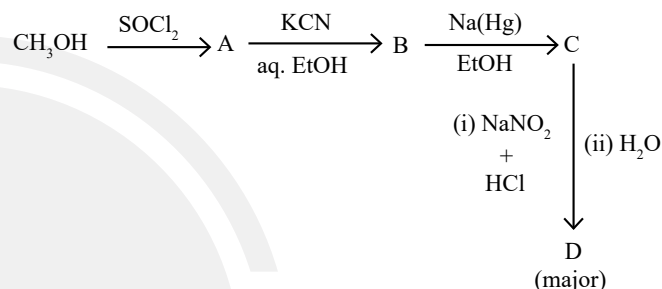
	List-I (Reaction)		List-II (Product formed)
A.	Gabriel synthesis	I.	Benzaldehyde
B.	Kolbe synthesis	II.	Ethers
C.	Williamson synthesis	III.	Primary amines
D.	Etard reaction	IV.	Salicylic acid

Choose the correct answer from the options given below

- a. A-III, B-IV, C-II, D-I
b. A-III, B-I, C-II, D-IV
c. A-II, B-III, C-I, D-IV
d. A-IV, B-III, C-I, D-II
4. Reaction of propanamide with ethanolic sodium hydroxide and bromine will give (2020-Covid)
- a. Methylamine
b. Propylamine
c. Aniline
d. Ethylamine
5. Method by which Aniline cannot be prepared is: (2015 Re)
- a. Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
b. Hydrolysis of phenylisocyanide with acidic solution
c. Degradation of benzamide with bromine in alkaline solution
d. Reduction of nitrobenzene with H_2/Pd in ethanol

Physical & Chemical Properties of Amines

6. The major product D formed in the following reaction sequence is: (2024 Re)



7. The compound that does not undergo Friedel-Crafts alkylation reaction but gives a positive carbylamine test is: (2024 Re)

- a. Aniline
b. Pyridine
c. N-methylaniline
d. Triethylamine

8. Given below are two statements: (2024)

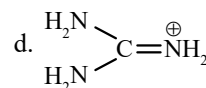
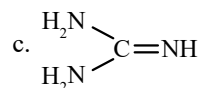
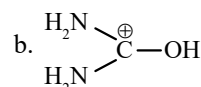
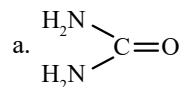
Statement-I: Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement-II: Aniline cannot be prepared through Gabriel synthesis.

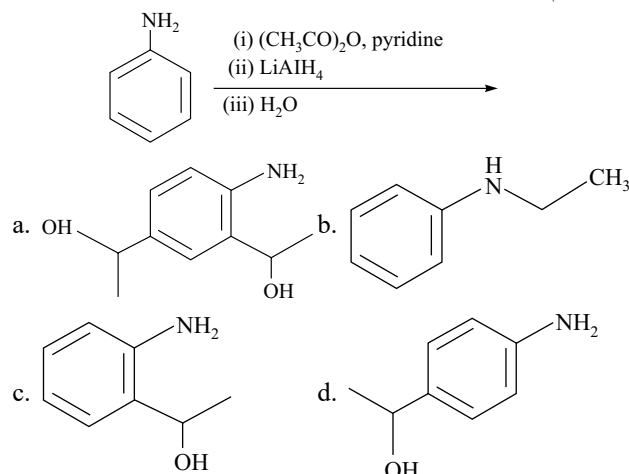
In the light of the above statements, choose the correct answer from the options given below:

- a. Statement-I is correct but Statement-II is false.
b. Statement-I is incorrect but Statement-II is true.
c. Both Statement-I and Statement-II are true.
d. Both Statement-I and Statement-II are false.

9. Which amongst the following compound/species is least basic? (2023-Manipur)



10. The product formed from the following reaction sequence is: (2022 Re)



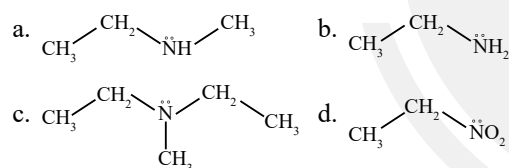
11. Given below are two statements: (2022)

Statement-I: Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.

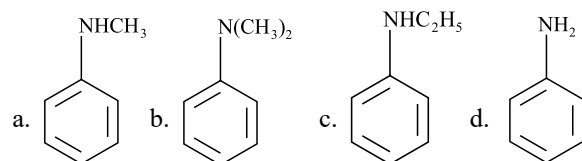
Statement-II: Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.

In the light of the above Statements, choose the **most appropriate** answer from the options given below:

- a. Statement-I is incorrect but Statement-II is correct.
 b. Both Statement-I and Statement-II are correct.
 c. Both Statement-I and Statement-II are incorrect.
 d. Statement-I is correct but Statement-II is incorrect.
12. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali. (2021)



13. Which of the following amine will give the carbylamine test? (2020)



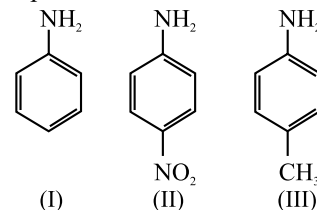
14. The correct order of the basic strength of methyl substituted amines in aqueous solution is: (2019)

- a. $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$
 b. $(\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH}$
 c. $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$
 d. $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$

15. Nitration of aniline in strong acidic medium also gives m-nitroaniline because: (2018)

- a. In spite of substituents nitro group always goes to only m-position.
 b. In electrophilic substitution reactions, amino group is meta directive.
 c. In acidic (strong) medium, aniline is present as anilinium ion.
 d. In absence of substituents nitro group always goes to m-position.

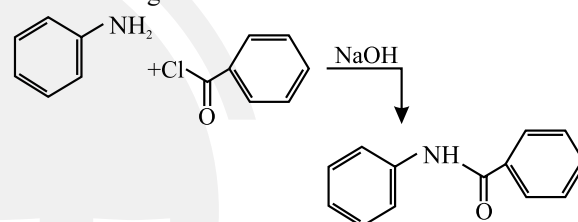
16. The correct increasing order of basic strength for the following compounds is: (2017-Delhi)



- a. $\text{II} < \text{I} < \text{III}$
 b. $\text{II} < \text{III} < \text{I}$
 c. $\text{III} < \text{I} < \text{II}$
 d. $\text{III} < \text{II} < \text{I}$
17. The correct statement regarding the basicity of arylamines is: (2016 - I)

- a. Arylamines are generally more basic than alkylamines, because the nitrogen atom in arylamines is sp -hybridised.
 b. Arylamines are generally less basic than alkylamines because the nitrogen lone pair electrons are delocalized by interaction with the aromatic ring π electrons system.
 c. Arylamines are generally more basic than alkylamines because the nitrogen lone pair electrons are not delocalised by interaction with the aromatic ring π electron system.
 d. Arylamines are generally more basic than alkylamines because of aryl group

18. The following reaction



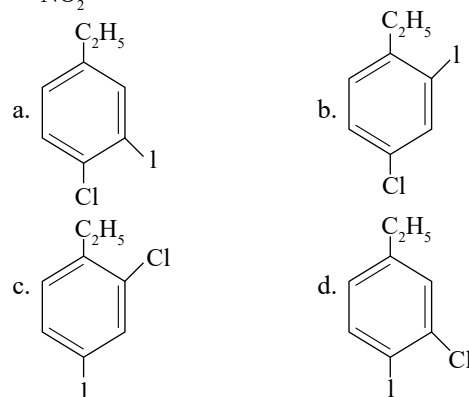
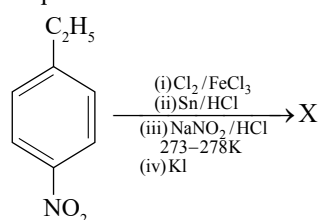
is known by the name:

- a. Schotten-Baumen reaction
 b. Friedel-Craft's reaction
 c. Perkin's reaction
 d. Acetylation reaction

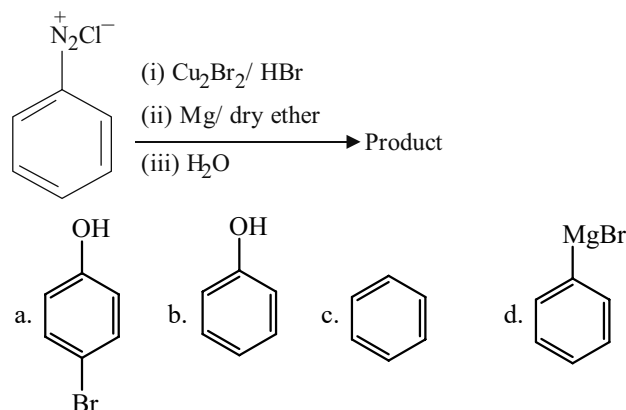
[OS] (2015 Re)

Diazonium Salt (Preparation, Physical & Chemical Properties)

19. The major product X formed in the following reaction sequence is: (2024 Re)



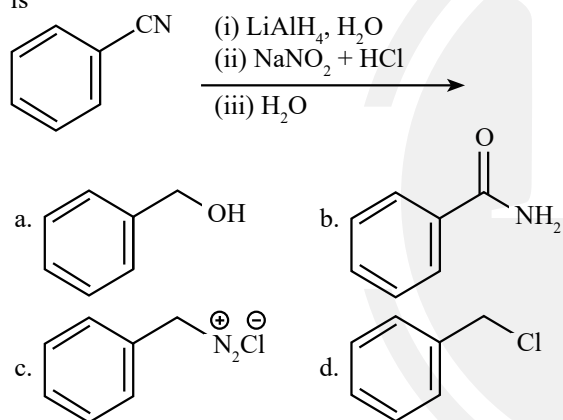
20. Identify the product in the following reaction: (2023)



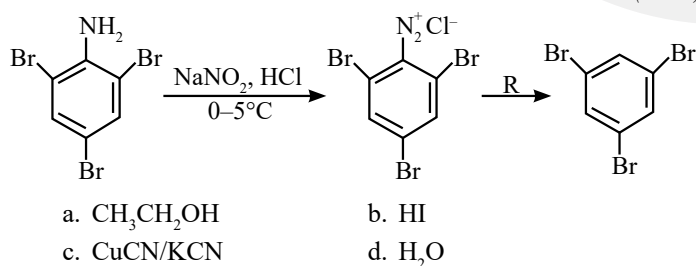
21. Choose the **correct** sequence of reagents in the conversion of 4-nitrotoluene to 2-bromotoluene. (2023-Manipur)

- a. $\text{NaNO}_2/\text{HCl}; \text{Sn}/\text{HCl}; \text{Br}_2; \text{H}_2\text{O}/\text{H}_3\text{PO}_2$
 b. $\text{Sn}/\text{HCl}; \text{NaNO}_2/\text{HCl}; \text{Br}_2; \text{H}_2\text{O}/\text{H}_3\text{PO}_2$
 c. $\text{Br}_2; \text{Sn}/\text{HCl}; \text{NaNO}_2/\text{HCl}; \text{H}_2\text{O}/\text{H}_3\text{PO}_2$
 d. $\text{Sn}/\text{HCl}; \text{Br}_2; \text{NaNO}_2/\text{HCl}; \text{H}_2\text{O}/\text{H}_3\text{PO}_2$

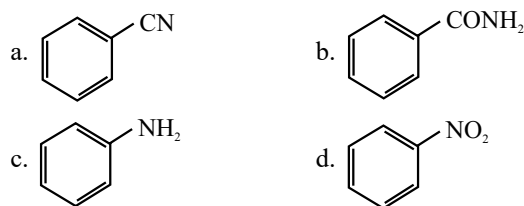
22. The product formed from the following reaction sequence is (2022)



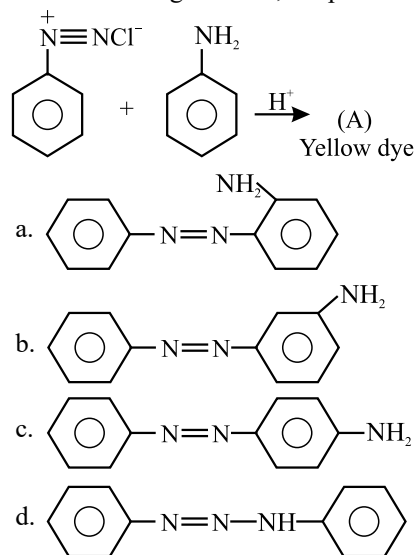
23. The reagent 'R' in the given sequence of chemical reaction is: (2021)



24. A given nitrogen-containing aromatic compound A reacts with Sn/HCl , followed by HNO_2 to give an unstable compound B. B, on treatment with phenol, forms a beautiful coloured compound C with the molecular formula $\text{C}_{12}\text{H}_{10}\text{N}_2\text{O}$. The structure of compound A is: (2016 - II)



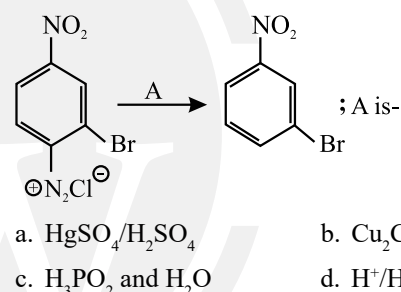
25. In the following reaction, the product (A) is: (2014)



26. Which of the following will be most stable diazonium salt RN_2^+X^- ? (2014)

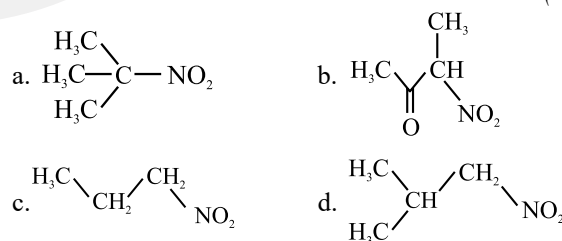
- a. $\text{C}_6\text{H}_5\text{N}_2^+\text{X}^-$ b. $\text{CH}_3\text{CH}_2\text{N}_2^+\text{X}^-$
 c. $\text{C}_6\text{H}_5\text{CH}_2\text{N}_2^+\text{H}^-$ d. $\text{CH}_3\text{N}_2^+\text{X}^-$

27. In the reaction: (2013)



Nitrobenzene (Preparation and Properties)

28. Which one of the following nitro-compounds does not react with nitrous acid? (2016 - II)



29. The electrolytic reduction of nitrobenzene in strongly acidic medium produces (2015)

- a. Azobenzene b. Aniline
 c. p-aminophenol d. Azoxybenzene

30. Nitrobenzene on reaction with conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$ at $80^\circ - 100^\circ\text{C}$ forms which one of the following products? (2013)

- a. 1,2-Dinitrobenzene
 b. 1,3-Dinitrobenzene
 c. 1,4-Dinitrobenzene
 d. 1,2,4-Trinitrobenzene

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (d) | 3. (a) | 4. (d) | 5. (a) | 6. (c) | 7. (a) | 8. (c) | 9. (b) | 10. (b) |
| 11. (d) | 12. (b) | 13. (d) | 14. (a) | 15. (c) | 16. (a) | 17. (b) | 18. (a) | 19. (c) | 20. (c) |
| 21. (c) | 22. (a) | 23. (a) | 24. (d) | 25. (c) | 26. (a) | 27. (c) | 28. (a) | 29. (c) | 30. (b) |

